MIT 18.06

Lecture 2 : Elimination with Matrices

→ The goal of this lecture is to transfer $Ax = b$ into Upper Triangular System $Ux = C$

<u>Upper Triangular Matrix</u> : It has zeroes everywhere below the main diagonal, making it easy to solve using back substitution.

<u>Let's take an example of Elimination</u> :

$$x + 2y + z = 2$$
$$3x + 8y + z = 12$$
$$4y + z = 2$$

Firstly convert it into Augmented Matrix :

Multiply row1 and subtract from row2

Multiply row2 and subtract from row3

$$\left[\begin{array}{ccc|c} \boxed{1} & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} \boxed{1} & 2 & 1 & 2 \\ 0 & \boxed{2} & -2 & 6 \\ 0 & 4 & 1 & 2 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} \boxed{1} & 2 & 1 & 2 \\ 0 & \boxed{2} & -2 & 6 \\ 0 & 0 & \boxed{5} & -10 \end{array}\right]$$

pivot

$$\underbrace{\qquad\qquad}_{U} \quad \underbrace{\qquad}_{C}$$

equation we get :

$$x + 2y + z = 2$$
$$2y - 2z = 6$$
$$5z = -10$$

After doing back substitution we get :

$$x = 2$$
$$y = 1$$
$$z = -2$$

<u>Elementary matrix</u> : (E_{ij}) The matrix used to perform a row operation.

So, subtract $3 \times$ row1 from row2

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{21}} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}$$

Subtract $2 \times$ row2 from row3

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}}_{E_{32}} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}$$

$$E_{32}(E_{21}A) = U$$
$$(E_{32}E_{21})A = U$$

<u>Augmented matrix</u> : It is a shorthand way of representing a linear equation using numbers.

<u>Pivot</u> : The first non-zero entry in each row as you eliminate. (Pivot $\neq 0$)

<u>Matrix Multiplication Rule</u> : Row operation are performed : $E_{21}A = U$

Sucess v/s failure

1) Sucess :- n pivot for an $n \times n$ matrix
2) Temporary failure :- A zero appear in pivot position
   - fix → Row exchange (swap with a row below)
3) Permanent failure :- A zero appear in pivot position and there are non zero entries below it.

   Result → The matrix is singular (non-invertible)

Permutation matrix :-

If we need to swap rows to move a pivot into place, we use a Permutation Matrix.

◎ Let's take example!

If I want to exchange row 1 and row 2 in this :-

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

We use identity matrix of 2X2 :-

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

Rows are swap!

⇒ If we have two matrix A and B then,

$AB \neq BA$